Lecture 15 2025-10-21

Corollary: Polynomial Interpolation is unique
(R=0 when f is a polynomial of deg $\leq n$)

Reason: $f^{(n+1)}(\xi) = 0$ if f is a polynomial of degree $\leq n$

Say P & $\tilde{P}$ both interpolate a function f at $x_0, \dots, x_n$
then the difference $D = P - \tilde{P}$ is a polynomial of degree $\leq n$

$$D(x) = (P(x) - \tilde{P}(x)) + R(x) - \tilde{R}(x)$$

$$R(x) = \frac{f^{(n+1)}(\xi(x))}{(n+1)!}(x-x_0)\cdots(x-x_n)$$

$= \tilde{R}(x)$ (same interpolation points and zeros of g)

$D(x)$ has $n+1$ zeros but has degree less than n
$\Rightarrow$ it must be identical to zero.

Example: $f(x) = \cos\left(\frac{\pi x}{2}\right)$ approximate by $P(x) = 1 - x^2$
on $[-1, 1]$ with $x_0 = -1$, $x_1 = 0$ & $x_2 = 1$

$\cos(0) = 1$, $\cos\left(-\frac{\pi}{2}\right) = 0 = \cos\left(\frac{\pi}{2}\right)$

$P(0) = 1$, $P(\pm 1) = 0$

$\Rightarrow$ P interpolates $\cos\left(\frac{\pi x}{2}\right)$ at x_0, x_1 & x_2

Solution: $R(x) = \frac{f^{(3)}(\xi)}{3!} \underbrace{(x+1)x(x-1)}_{:= r(x)}$

$$|f^{(3)}(\xi)| = \left|\left(\frac{\pi}{2}\right)^3 \sin\left(\frac{\pi\xi}{2}\right)\right| \leq \left(\frac{\pi}{2}\right)^3$$

$r(x) = x^3 - x$ $\qquad r(-1) = r(1) = 0$

$r'(x) = 3x^2 - 1 \overset{!}{=}$ $x = \pm\sqrt{\frac{1}{3}}$ are extrema

$$r\left(\pm\sqrt{\tfrac{1}{3}}\right) = \mp \frac{2}{3\sqrt{3}}$$

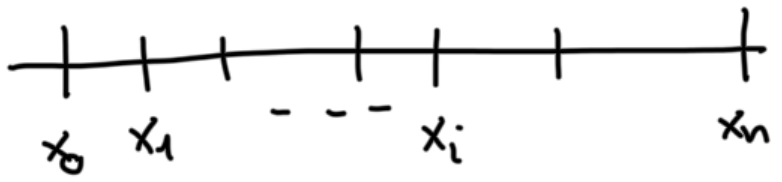
$$|R(x)| \leq \frac{1}{6}\cdot\frac{\pi^3}{8}\cdot\frac{2}{3\sqrt{3}} \approx 0.2486 \qquad \text{actual error } \sim 0.0561$$

Taylor's Polynomial $P_3(x) = 1 - \frac{\pi^2}{8}x^2$ $\qquad |R_3(x)| = \left|\frac{f^{(4)}(\xi)}{4!}x^4\right|$

≤ 0.2537

3.2 Recursive Interpolation

n+1 points



Notation: $P(x)$ interpolates $f(x)$ at $x_0, \ldots, x_n$

$P_{4:7}(x)$ interpolates $f(x)$ at x_4, x_5, x_6 & x_7

$P_{i:j}(x)$ interpolates $f(x)$ at $x_i, \ldots, x_j$

$\hat{P}_k(x)$ interpolates $f(x)$ at all x_i except $i=k$

typo in lecture

Theorem: If $i \neq j$, $P(x) = \dfrac{(x-x_j)\hat{P}_j(x) - (x-x_i)\hat{P}_i(x)}{x_i - x_j}$

Proof: We need to check that $P(x_k) = f(x_k) \quad \forall\ k = 0, \dots, n$

$$P(x_i) = f(x_i) = \frac{(x_i - x_j)\hat{P}_j(x_i)}{x_i - x_j} = \hat{P}_j(x_i) = f(x_i)$$

$$P(x_j) = \frac{-(x_j - x_i)\hat{P}_i(x_j)}{x_j - x_j} = \hat{P}_i(x_j) = f(x_j)$$

$k \neq i, j$

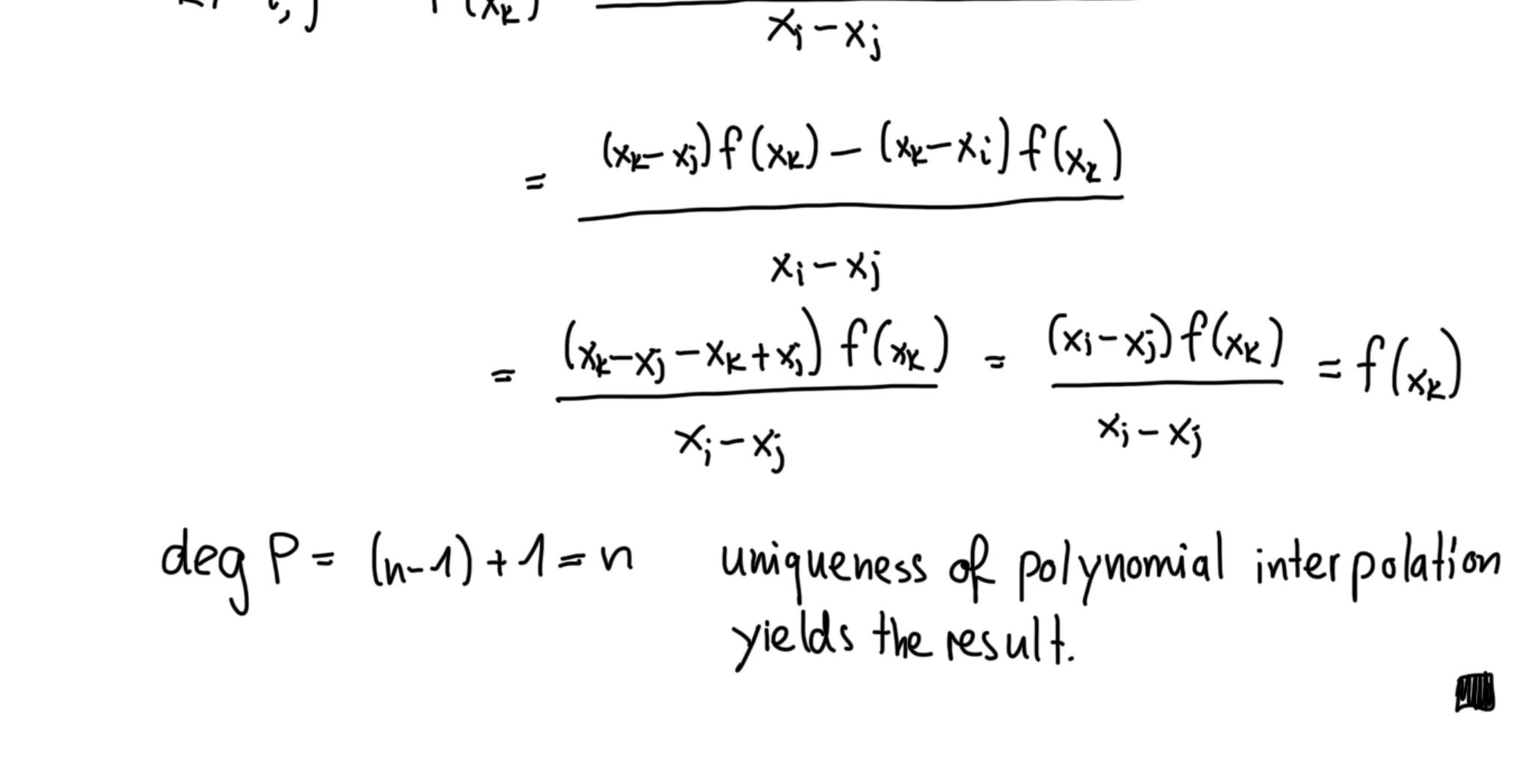
$$P(x_k) = \frac{(x_k - x_j)\hat{P}_j(x_k) - (x_k - x_i)\hat{P}_i(x_k)}{x_i - x_j}$$

$$= \frac{(x_k - x_j)f(x_k) - (x_k - x_i)f(x_k)}{x_i - x_j}$$

$$= \frac{(x_k - x_j - x_k + x_i)f(x_k)}{x_i - x_j} = \frac{(x_i - x_j)f(x_k)}{x_i - x_j} = f(x_k)$$

$\deg P = (n-1) + 1 = n$ uniqueness of polynomial interpolation yields the result. ∎

$\Rightarrow$ We can generate interpolating polynomials recursively



deg = 4

$P_{0:4} = P$

Make a table of polynomials interpolating at consecutive points

$$Q_{ij}(x) = P_{i-j:i}(x) = \frac{(x - x_{i-j})\,Q_{i,j-1}(x) - (x - x_i)\,Q_{i-1,j-1}(x)}{x_i - x_{i-j}}$$

($P_{i-j+1:i}$, $P_{i-j,i-1}$)

x fixed evaluation point

We build a hierarchy of polynomial approximations of increasing degree.

x_0 Q_{00}

x_1 Q_{10} Q_{11}

x_2 Q_{20} Q_{21} Q_{22}

x_3 Q_{30} Q_{31} Q_{32} Q_{33}

x_4 Q_{40} Q_{41} Q_{42} Q_{43} Q_{44}

Examples: $Q_{20}(x) = P_{2:2}(x) = f(x_2)$

$$Q_{32}(x) = \frac{(x - x_1)Q_{31}(x) - (x - x_3)Q_{21}(x)}{x_3 - x_1}$$

$$= \frac{x - x_1}{x_3 - x_1} P_{2:3}(x) + \frac{x - x_3}{x_1 - x_3} P_{1:2}(x)$$

sum to 1 (weighted average)

Plug in x_1, x_2 & x_3 to see that Q_{32} interpolates f at x_1, x_2, x_3

Filling in the table involves a similar amount of work to computing $Q_{44}(x) = P_{0:4}(x) = f(x_0)L_0(x) + \ldots + f(x_4)L_4(x)$ but gives more information, e.g. if x is between x_1 and x_2 then

$Q_{21}(x) = P_{1:2}(x)$, $Q_{22}(x) = P_{0:2}(x)$, $Q_{32}(x) = P_{1:3}(x)$ and $Q_{33}(x) = P_{0:3}(x)$

are all approximations of various order & degree of accuracy.

⇒ Can compare numbers to estimate interpolation error.